

Homework 10

due 20.12.2023, 14:00

Exercise 10.1 - MPI + openMP Matrix-Matrix product (10 points)

Take your code from Exercise 6 which performed a parallel matrix-matrix multiplication relying on the "Allgather" of a matrix column and modify it accordingly such that the loops involved in the calculation of the product are distributed among different threads using openMP.

- The I/O and distribution of the matrix among processes should be handled in the same way as in Exercise 6. The first row of a matrix file contains m and n , the number of rows and columns of a matrix A . The remaining lines contain one matrix element each, in the order $a_{00}, a_{01}, \dots, a_{0n-1}, a_{10}, \dots, a_{m-1,n-1}$.
- To test your program, use the two matrices `A.txt` and `B.txt` that you can find in `/work/korzec/LAB2/ex7/` on stromboli (avoid making copies, these files can be accessed from the compute nodes for reading). These are square matrices. Your program should create the output `C.txt`, which can be visualized as an image with the matlab script `check.m`. Include this image in your PDF.

Run your code using 4 nodes, 1 MPI task per node such that all threads spawned per MPI task are in the same node. Time your code **without** including I/O using 1, 2, ..., 10 threads per MPI task and produce a strong scaling plot.